



南开大学
Nankai University

视觉语言模型CLIP的提示 学习方法研究

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Outlines

- Background
 - Vision-Language Models
 - Prompt Learning
 - Evaluation Metric
- Related Works
- [CVPR 24] PromptKD
- [ECCV 24] CasPL
- Conclusion

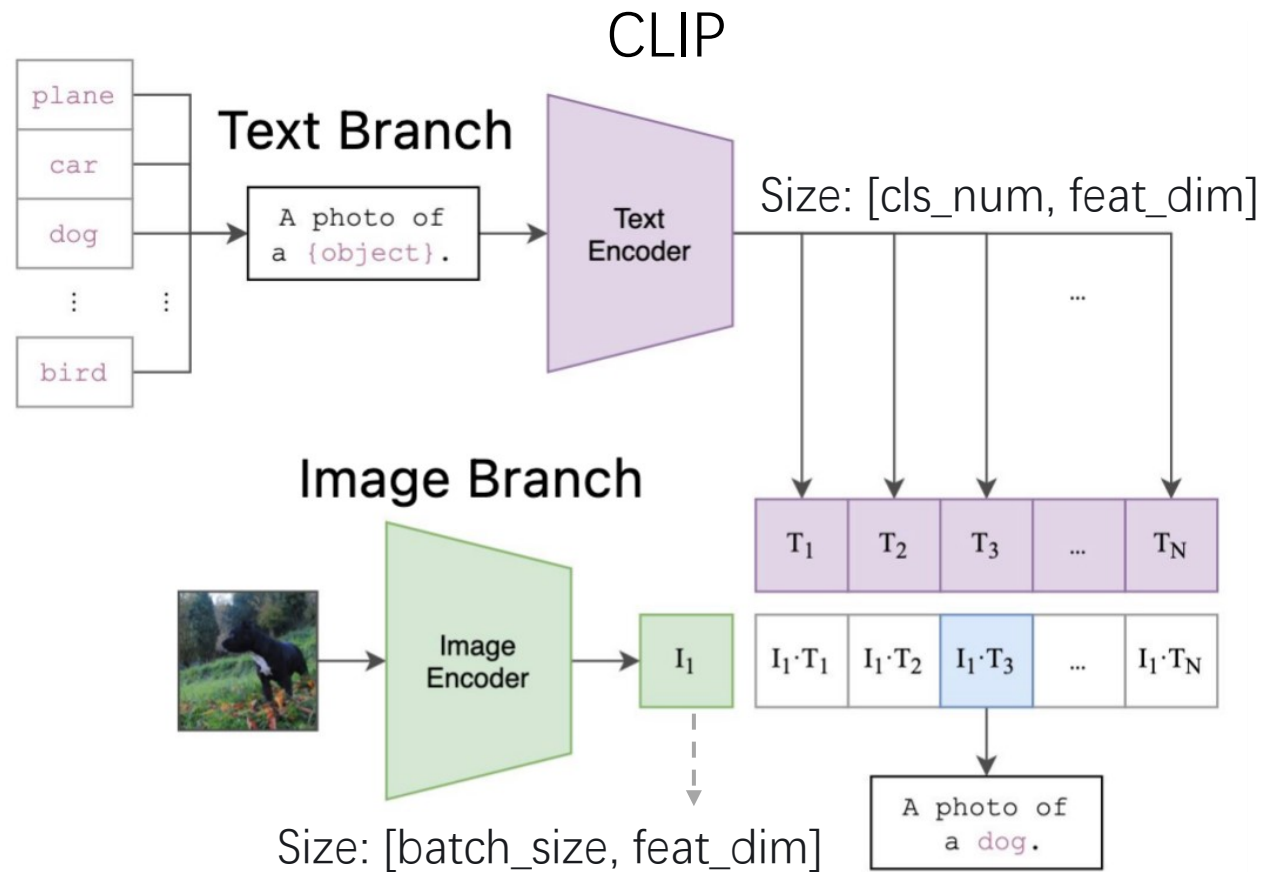


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What is vision-language model (CLIP)?



Training:
Contrastive loss + 400m image/text pairs.

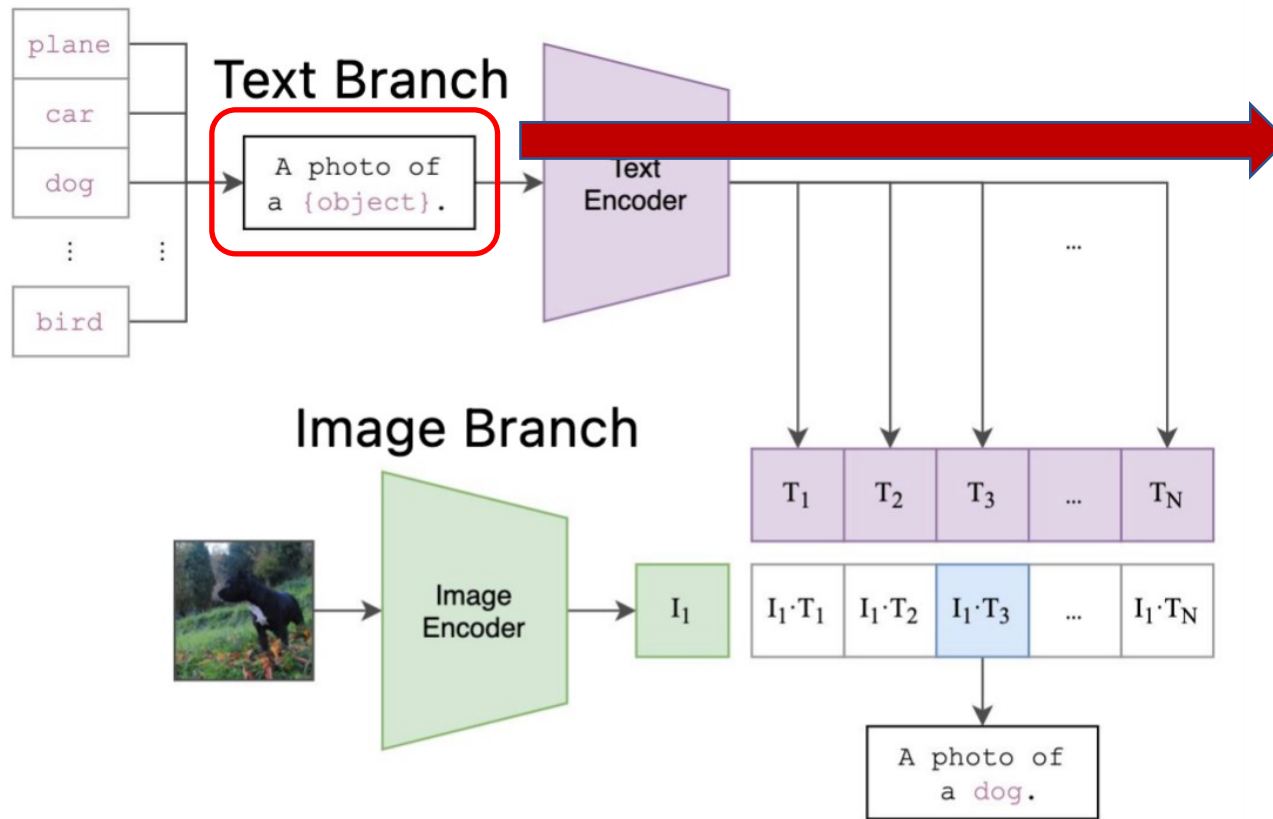
Advantages:

1. Prediction Flexibility.
2. Superior zero-shot performance.
3. Text features can be stored.



What is prompt learning?

Background:



Problems:

Fixed text prompts are suboptimal for various conditions.



For example...

Caltech101



Prompt	Accuracy
a [CLASS].	82.68
a photo of [CLASS].	80.81
a photo of a [CLASS].	86.29
$[V]_1 [V]_2 \dots [V]_M$ [CLASS].	91.83

(a)

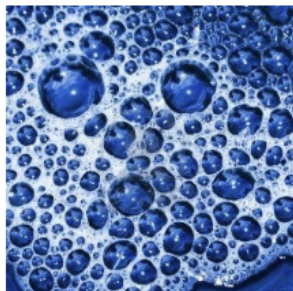
Flowers102



Prompt	Accuracy
a photo of a [CLASS].	60.86
a flower photo of a [CLASS].	65.81
a photo of a [CLASS], a type of flower.	66.14
$[V]_1 [V]_2 \dots [V]_M$ [CLASS].	94.51

(b)

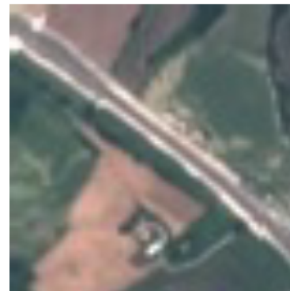
Describable Textures (DTD)



Prompt	Accuracy
a photo of a [CLASS].	39.83
a photo of a [CLASS] texture.	40.25
[CLASS] texture.	42.32
$[V]_1 [V]_2 \dots [V]_M$ [CLASS].	63.58

(c)

EuroSAT



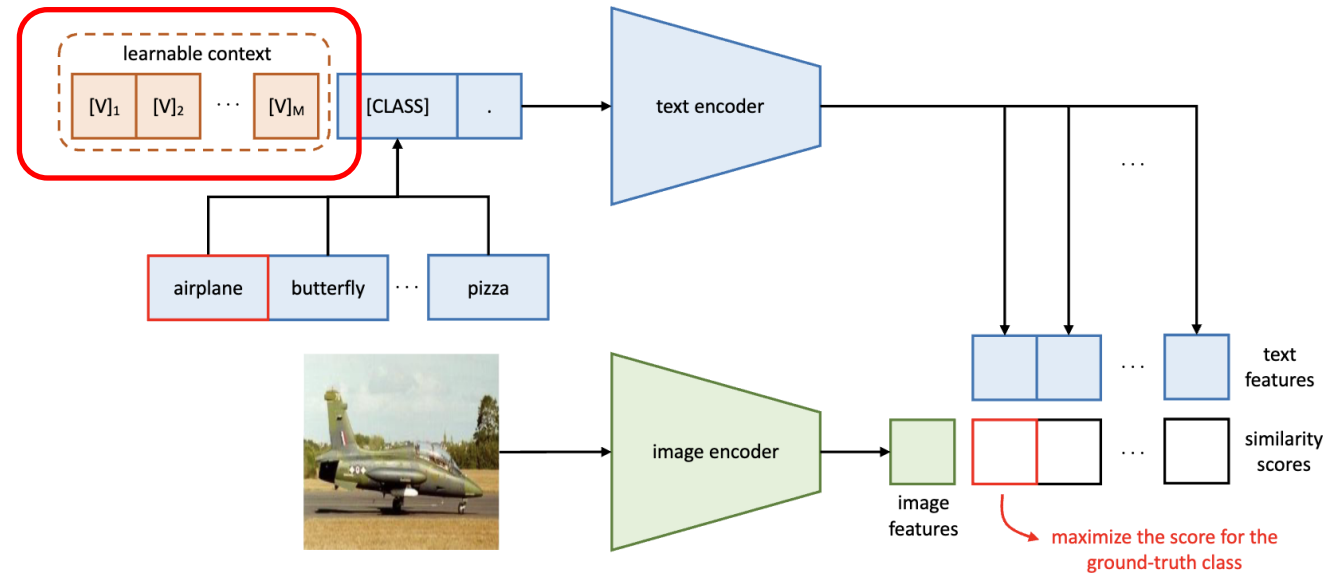
Prompt	Accuracy
a photo of a [CLASS].	24.17
a satellite photo of [CLASS].	37.46
a centered satellite photo of [CLASS].	37.56
$[V]_1 [V]_2 \dots [V]_M$ [CLASS].	83.53

(d)



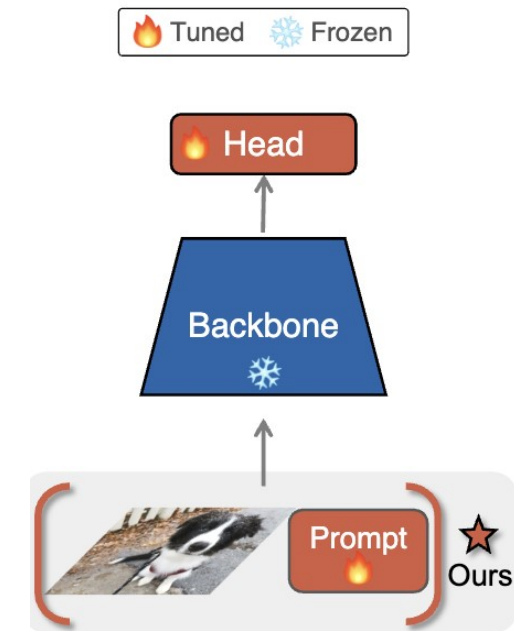
Prompt Learning

Text-based method:



CoOp

Image-based method:



VPT



Evaluation Metric

Table in PromptKD

ViT-B/16	Base	Novel	HM
CLIP	69.34	74.22	71.70
CoOp	82.69	63.22	71.66
CoCoOp	80.47	71.69	75.83
MaPLe	82.28	75.14	78.55
PromptSRC	84.26	76.10	79.97
PromptKD	86.96	80.73	83.73

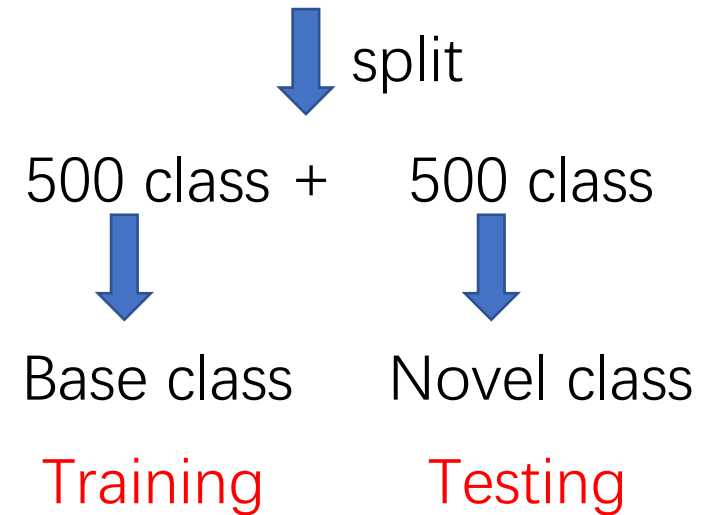
Base: Base Class Accuracy

Novel: Novel Class Accuracy

HM: Harmonic Mean

$$HM = (2 * \text{base} * \text{novel}) / (\text{base} + \text{novel})$$

For datasets like ImageNet-1K:



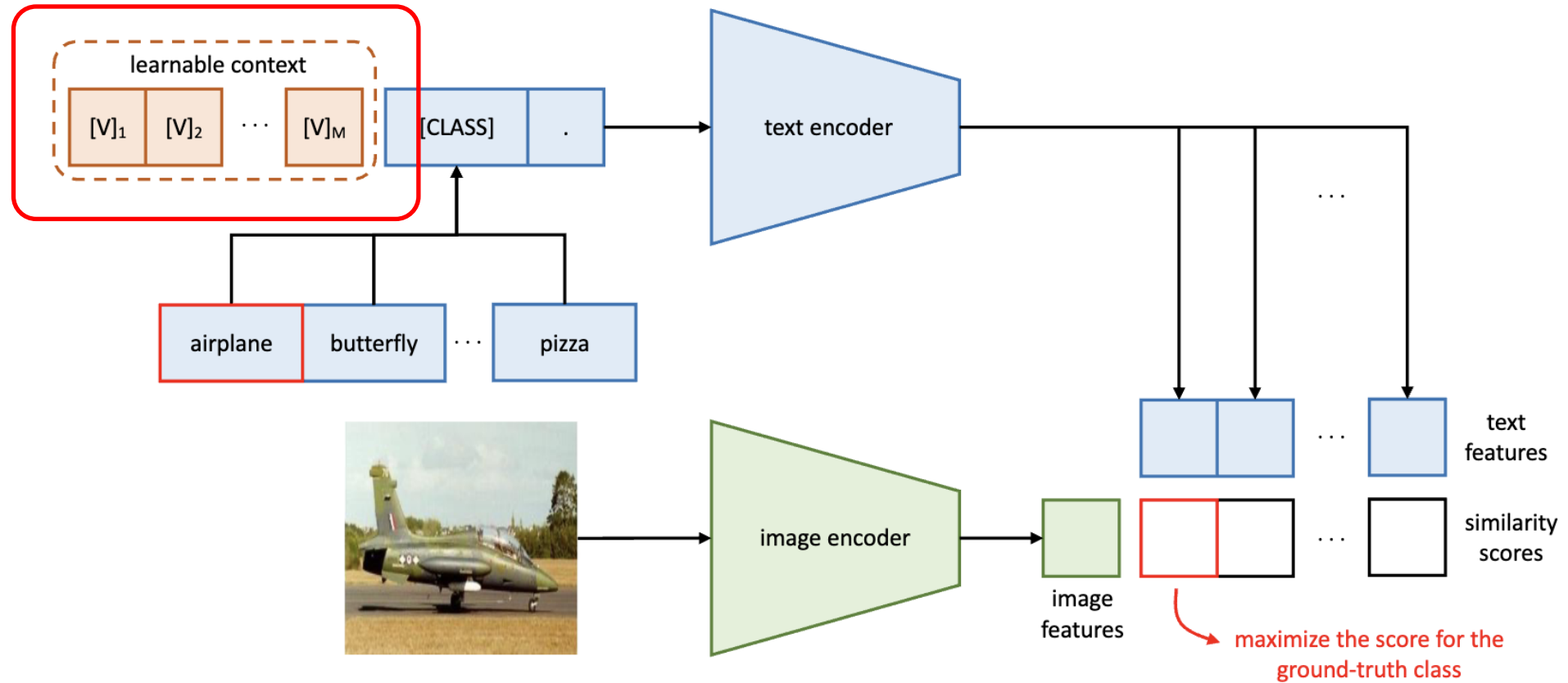
Outlines

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 - Introducing extra knowledge
 - Regularization
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[IJCV 22] CoOp

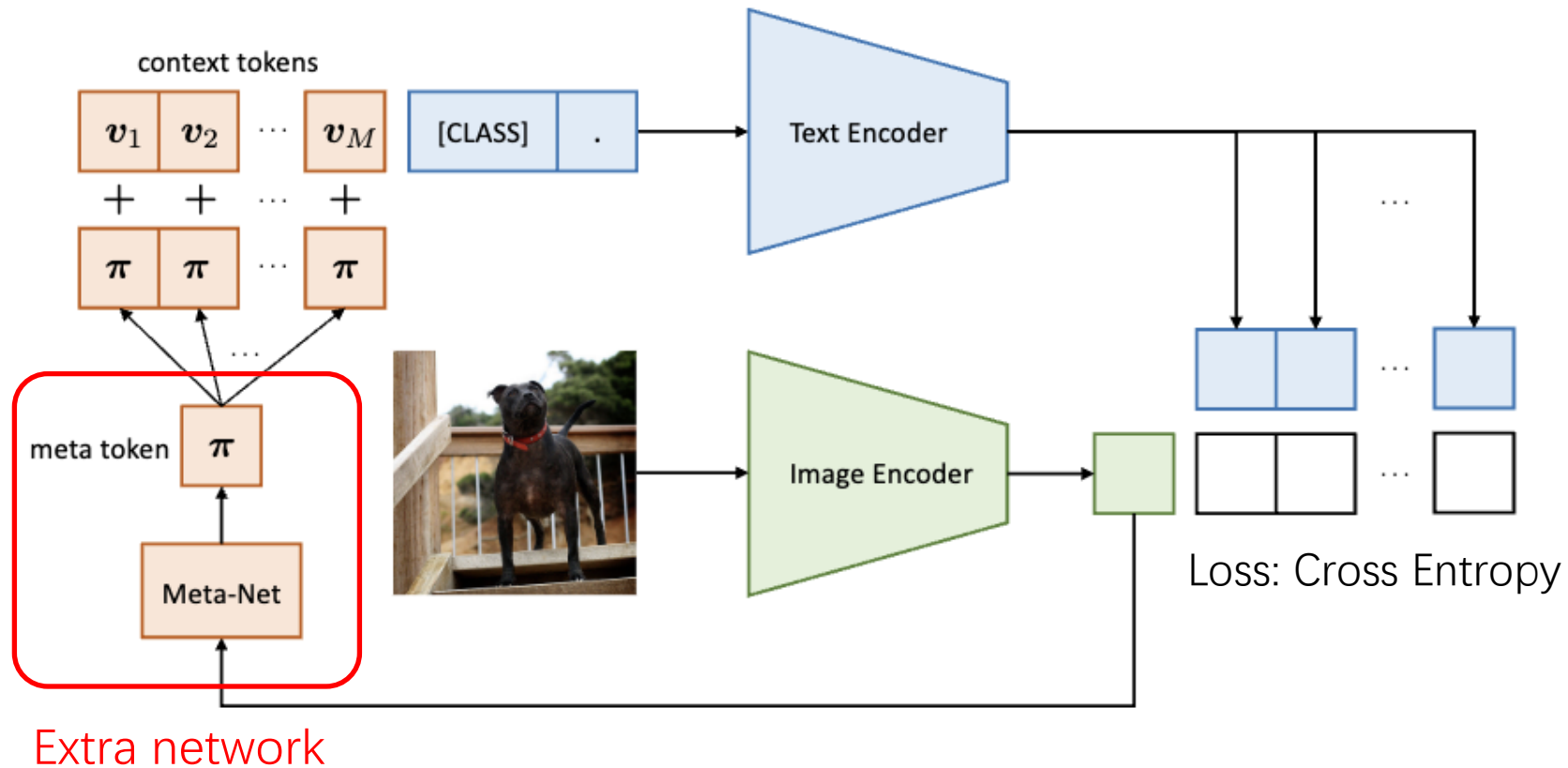
Learnable text tokens



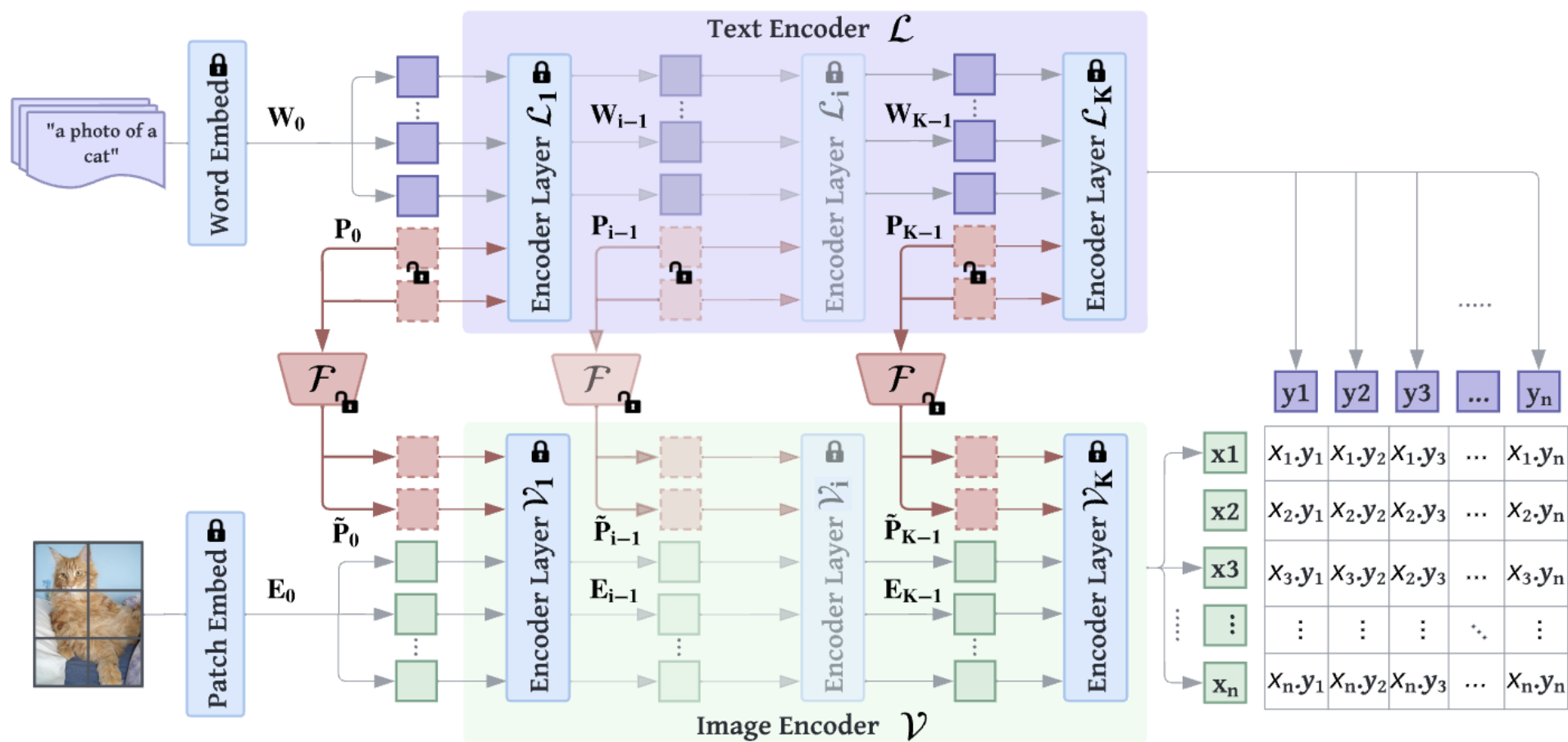
Loss: Cross Entropy



[CVPR 22] CoCoOp



[CVPR 23] MaPLE

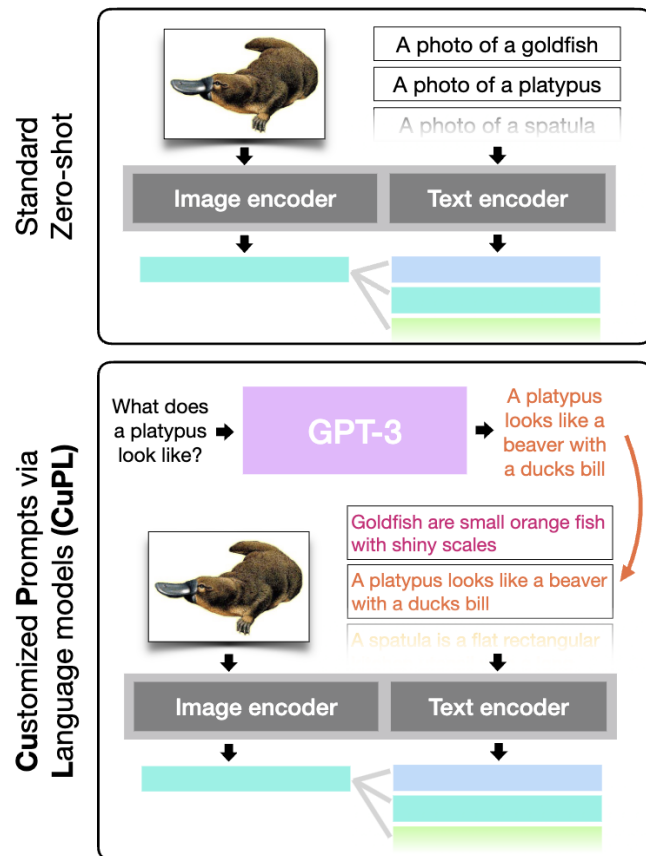


Bringing prompt learning to multimodality.

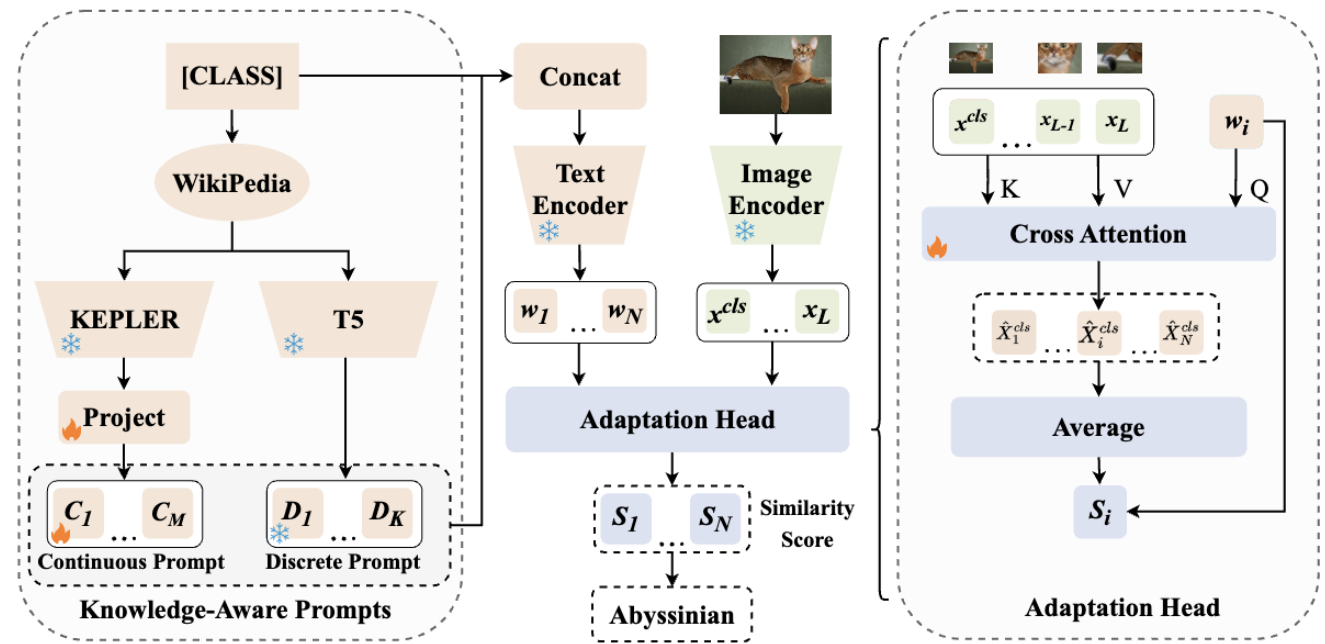


Introducing Extra Information

- Information source: wikipedia, LLMs, extra datasets...



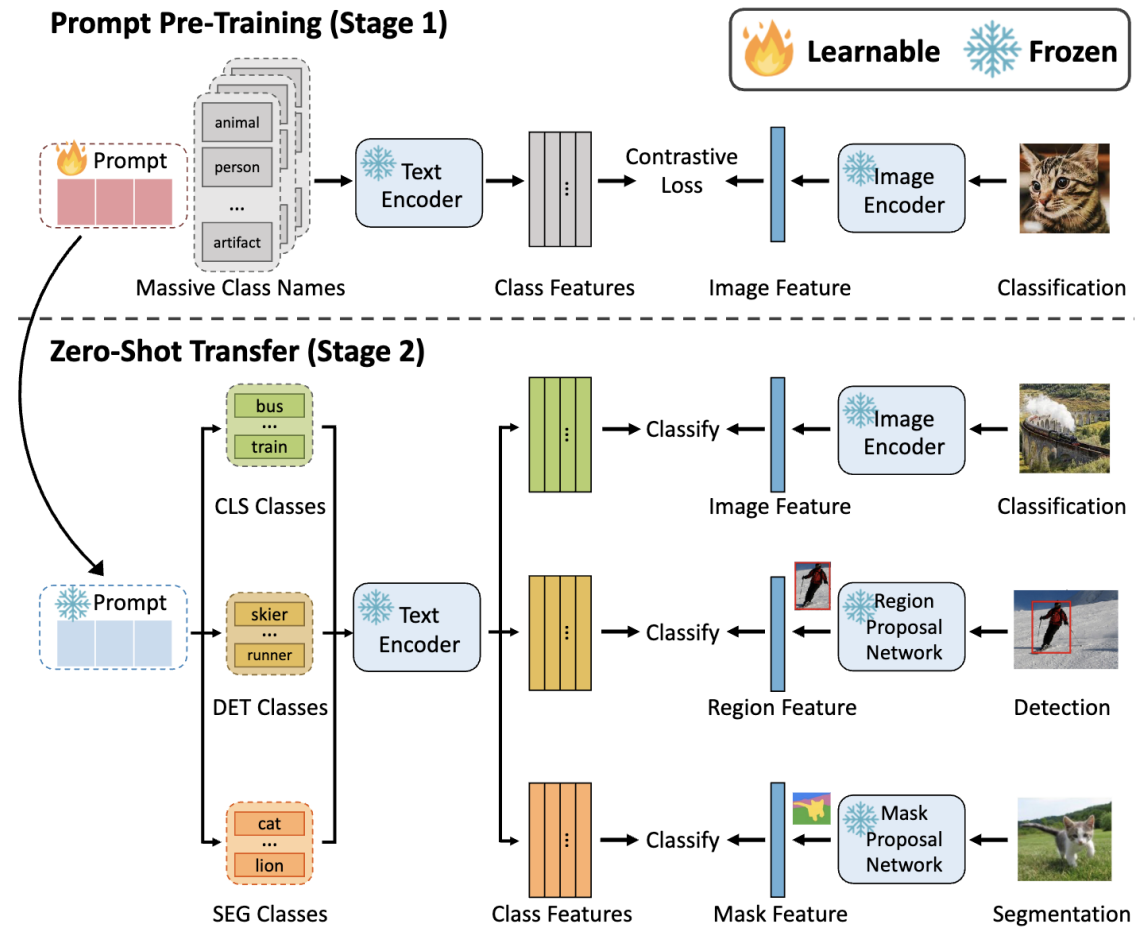
CuPL (ICCV 23)



KAPT (ICCV 23)



Extra Information: dataset

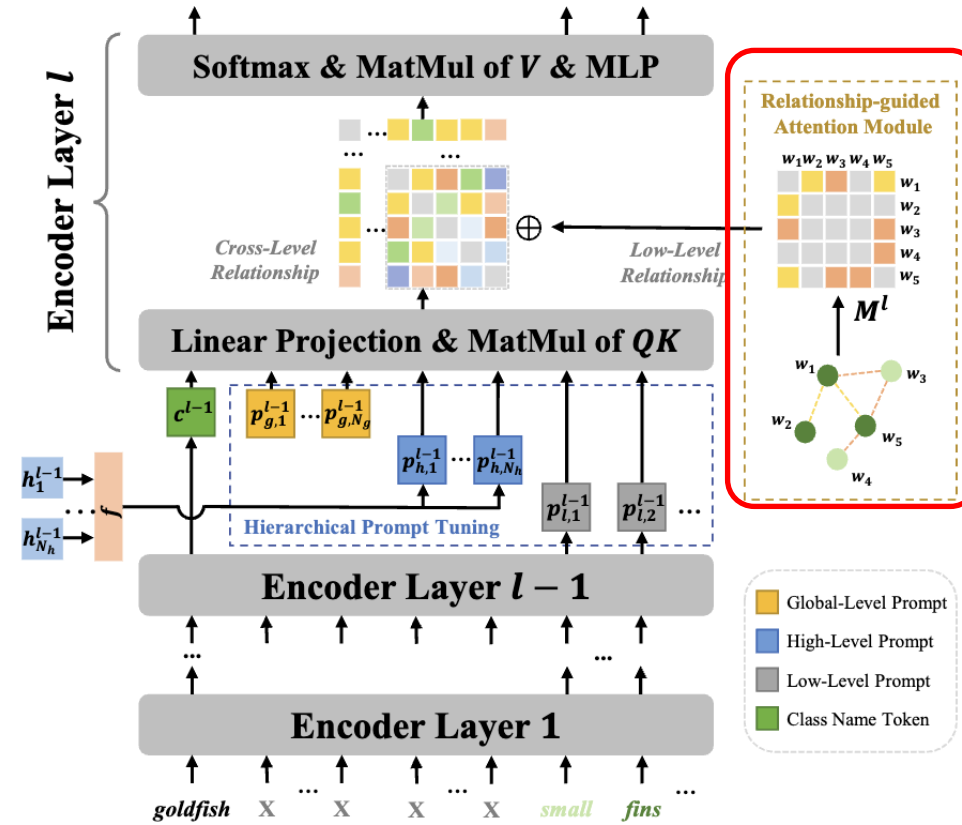
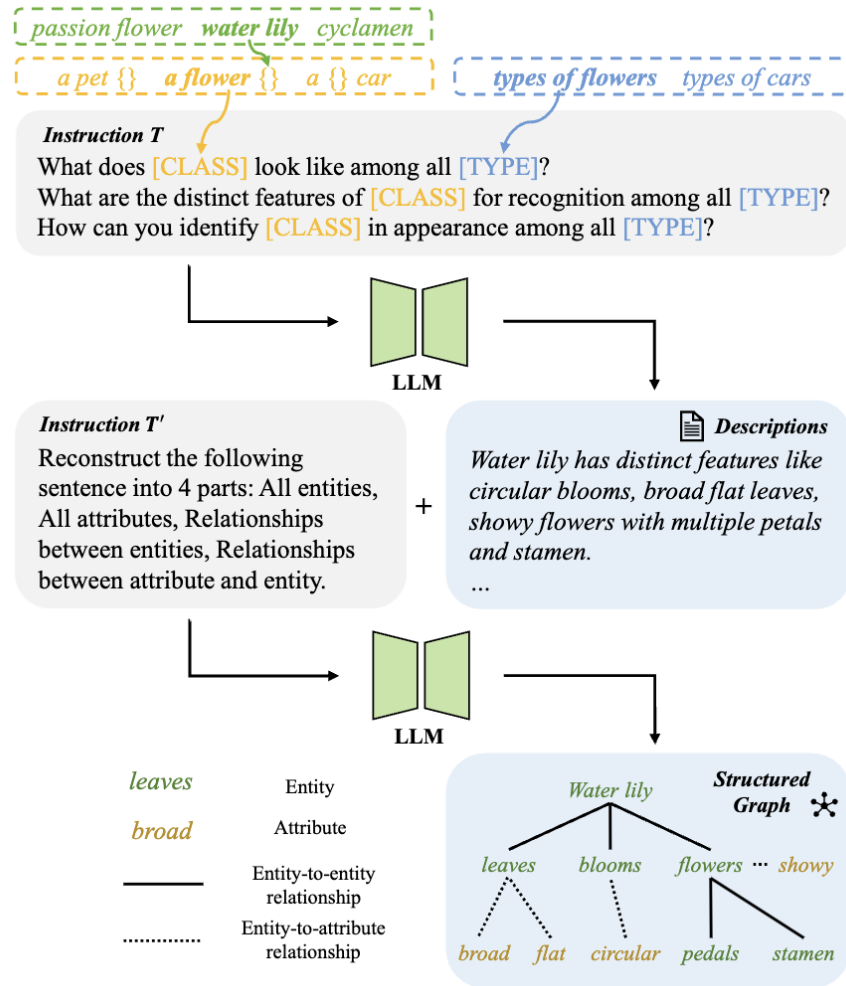


Dataset: ImageNet-21K

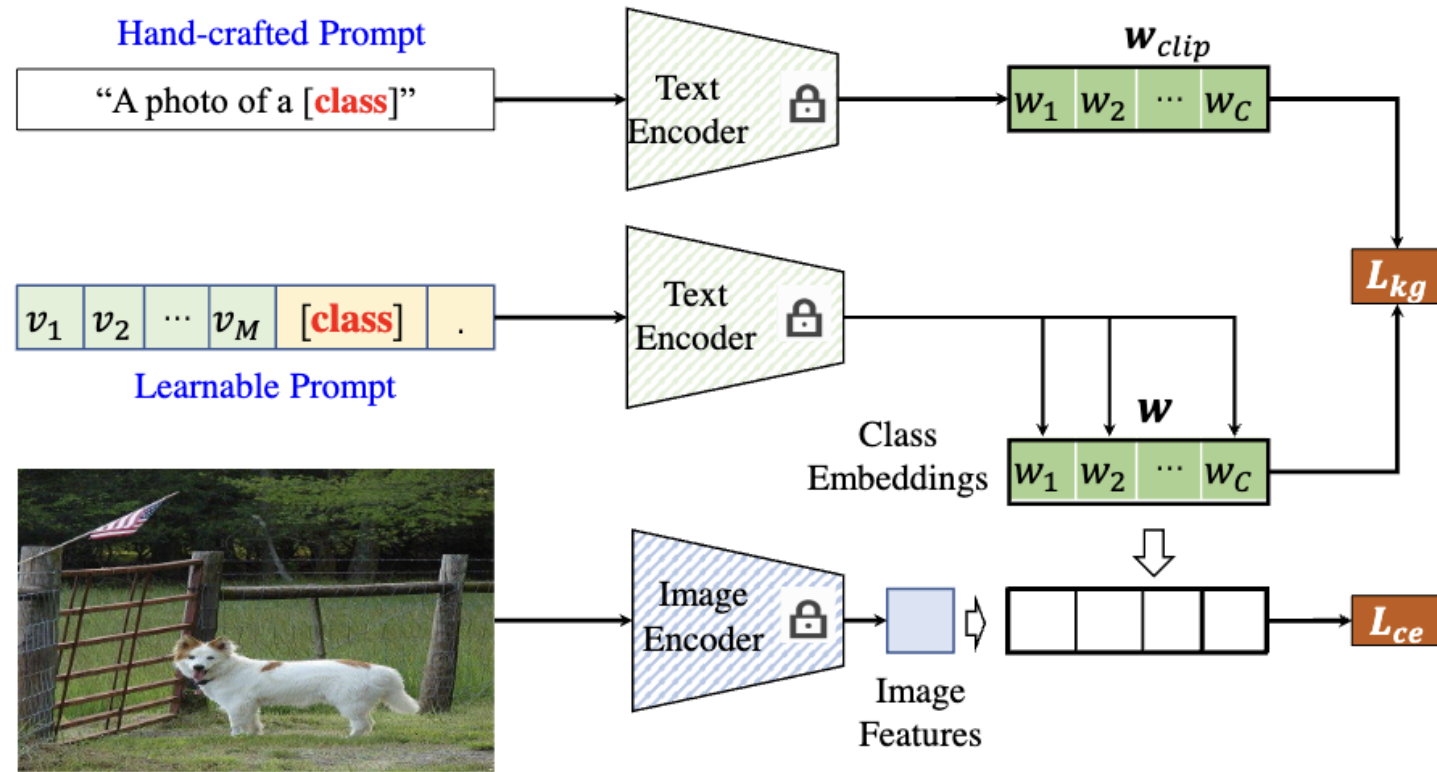
POMP (NeurIPS 2023)



Extra information: LLM & Graph model



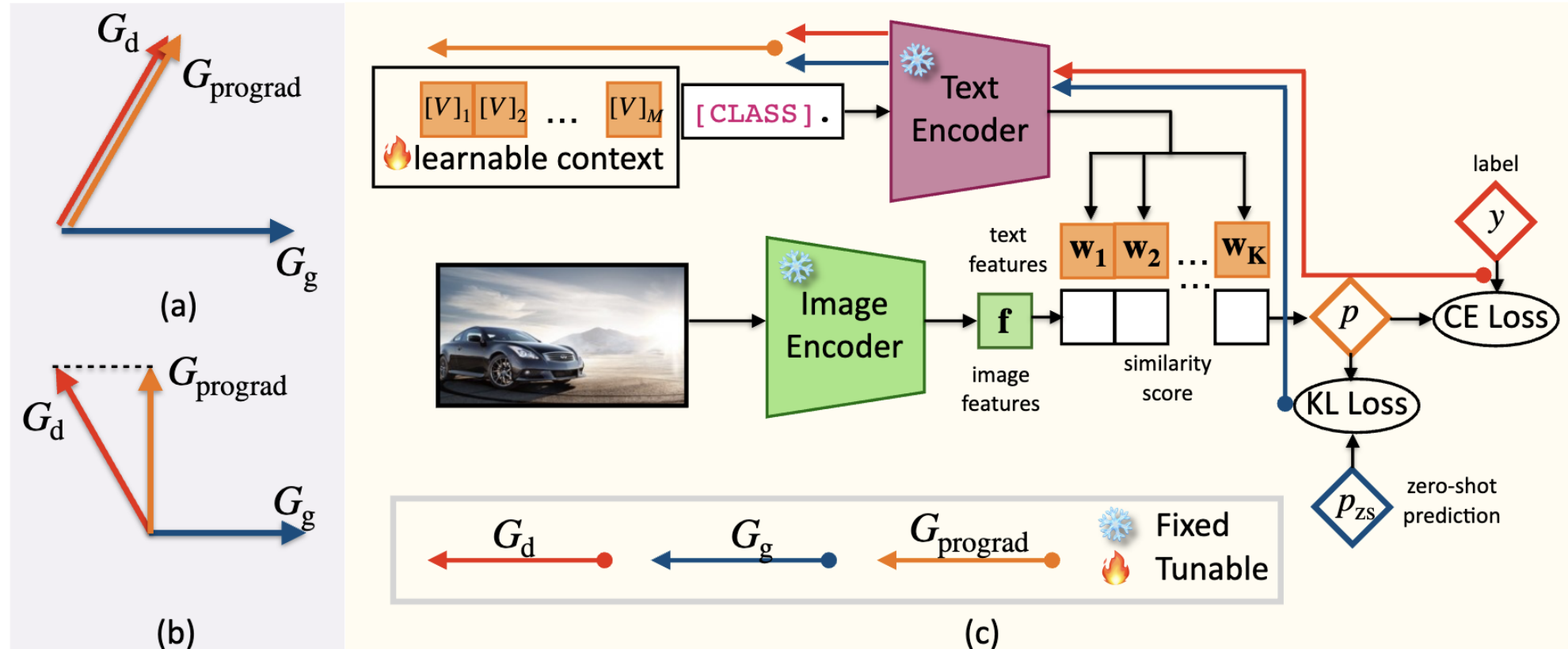
Regularization



KgCoOp (CVPR 23)



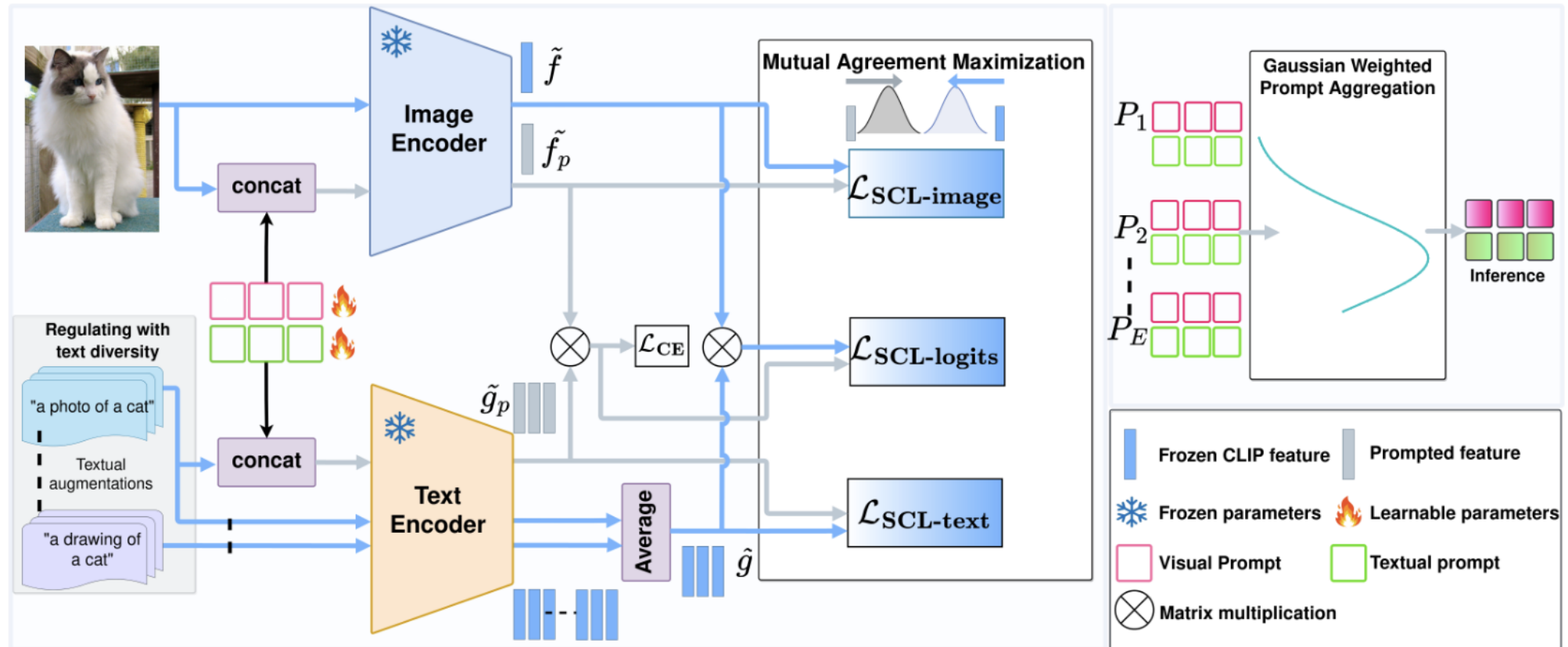
Regularization



ProGrad (ICCV 23)



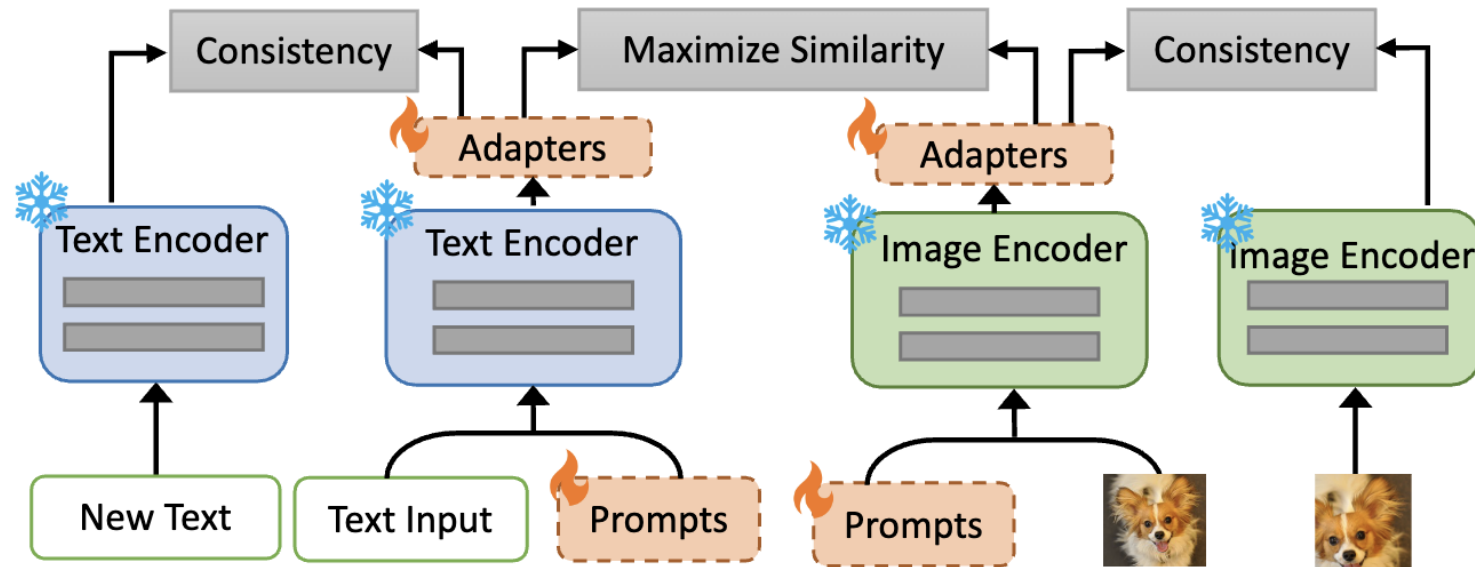
Regularization



PromptSRC (ICCV 23)



Regularization



(b) Our consistency-guided multimodal prompt tuning (CoPrompt)

Figure 1: Comparison between the CoPrompt and existing prompting approach.

CoPrompt (ICLR 24)



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PromptKD: Unsupervised Prompt Distillation for Vision-Language Models

Zheng Li, Xiang Li, Xinyi Fu, Xin Zhang, Weiqiang Wang, Shuo Chen, Jian Yang.

CVPR 2024

Email: zhengli97@mail.nankai.edu.cn

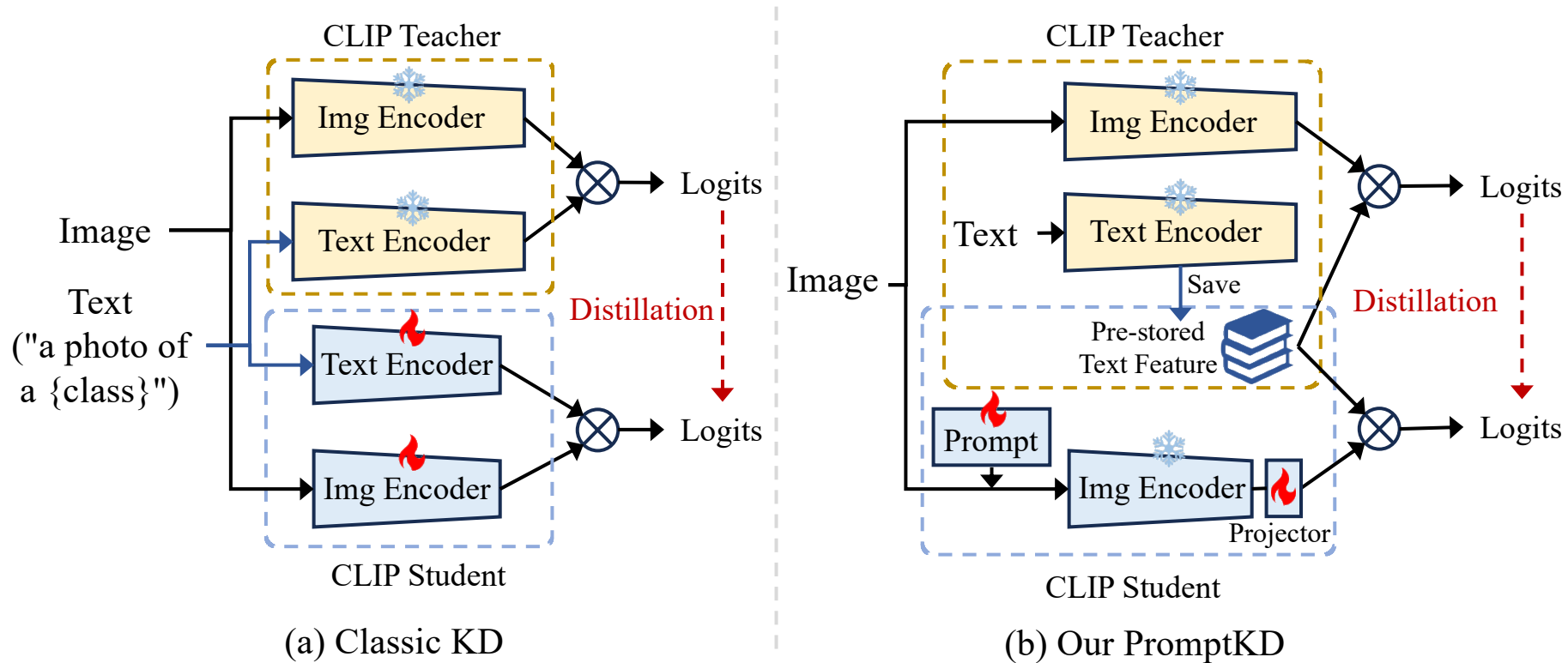
Code: <https://github.com/zhengli97/PromptKD>

Project Page: <https://zhengli97.github.io/PromptKD>

中文解读: <https://zhuanlan.zhihu.com/p/684269963>



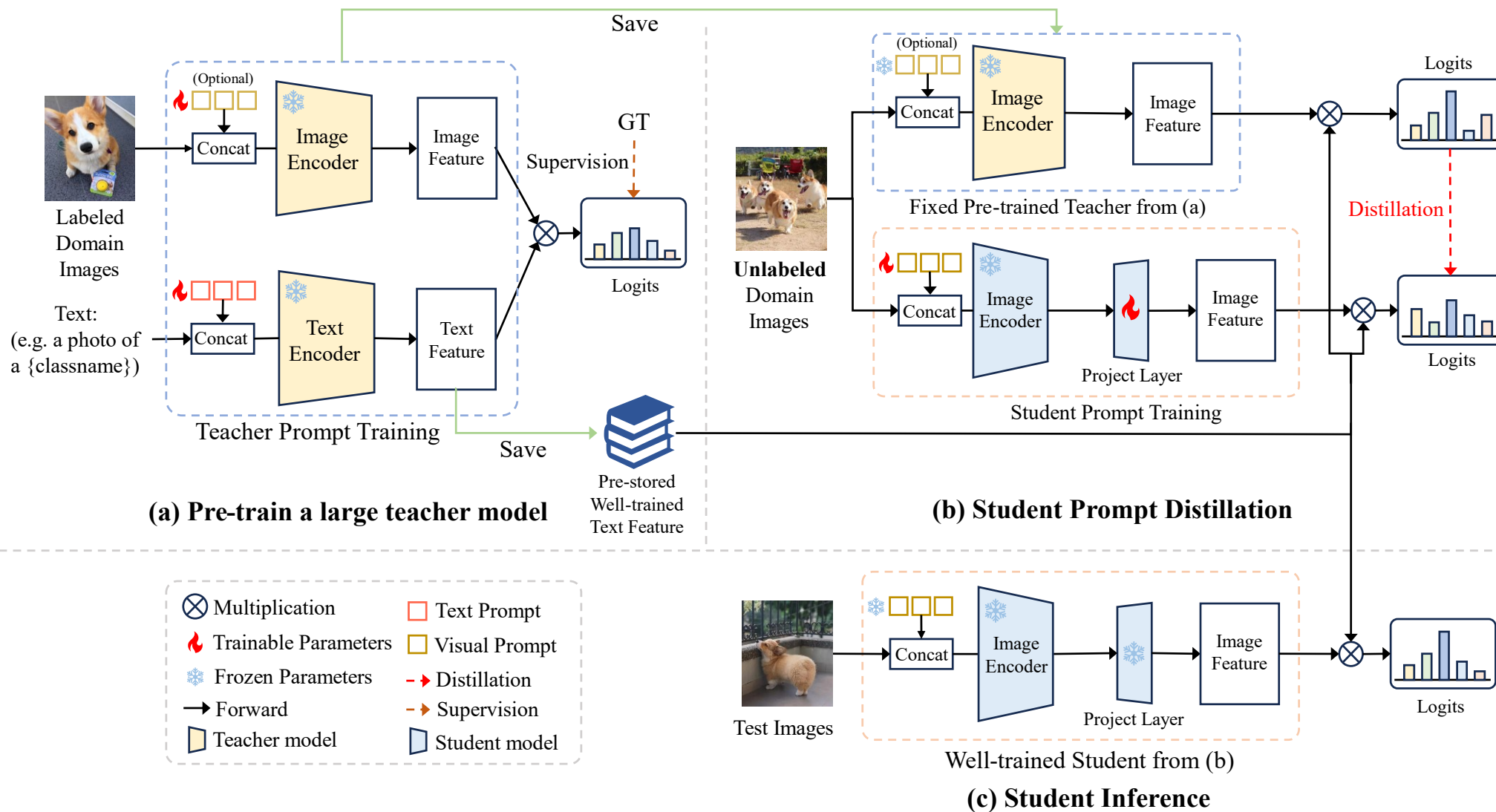
[CVPR 24] PromptKD



- Larger teacher models can provide better guidance for prompt learning.
- Leveraging decoupled-modality characteristics of CLIP can speed up training and inference.
- Training with extensive images leads to better prompt representations.



[CVPR 24] PromptKD



SOTA Performance

ViT-B/16	Base	Novel	HM
CLIP	69.34	74.22	71.70
CoOp	82.69	63.22	71.66
CoCoOp	80.47	71.69	75.83
MaPLe	82.28	75.14	78.55
PromptSRC	84.26	76.10	79.97
PromptKD	86.96	80.73	83.73
Δ	+2.70	+4.63	+3.76

(a) Average over 11 datasets.

ViT-B/16	Base	Novel	HM
CLIP	91.17	97.26	94.12
CoOp	93.67	95.29	94.47
CoCoOp	95.20	97.69	96.43
MaPLe	95.43	97.76	96.58
PromptSRC	95.33	97.30	96.30
PromptKD	96.30	98.01	97.15
Δ	+0.97	+0.71	+0.85

(d) OxfordPets

ViT-B/16	Base	Novel	HM
CLIP	90.10	91.22	90.66
CoOp	88.33	82.26	85.19
CoCoOp	90.70	91.29	90.99
MaPLe	90.71	92.05	91.38
PromptSRC	90.67	91.53	91.10
PromptKD	92.43	93.68	93.05
Δ	+1.76	+2.15	+1.95

(g) Food101

ViT-B/16	Base	Novel	HM
CLIP	53.24	59.90	56.37
CoOp	79.44	41.18	54.24
CoCoOp	77.01	56.00	64.85
MaPLe	80.36	59.18	68.16
PromptSRC	83.37	62.97	71.75
PromptKD	85.84	71.37	77.94
Δ	+2.47	+8.40	+6.19

(j) DTD

ViT-B/16	Base	Novel	HM
CLIP	72.43	68.14	70.22
CoOp	76.47	67.88	71.92
CoCoOp	75.98	70.43	73.10
MaPLe	76.66	70.54	73.47
PromptSRC	77.60	70.73	74.01
PromptKD	80.83	74.66	77.62
Δ	+3.23	+3.93	+3.61

(b) ImageNet

ViT-B/16	Base	Novel	HM
CLIP	63.37	74.89	68.65
CoOp	78.12	60.40	68.13
CoCoOp	70.49	73.59	72.01
MaPLe	72.94	74.00	73.47
PromptSRC	78.27	74.97	76.58
PromptKD	82.80	83.37	83.13
Δ	+4.53	+8.40	+6.55

(e) StanfordCars

ViT-B/16	Base	Novel	HM
CLIP	27.19	36.29	31.09
CoOp	40.44	22.30	28.75
CoCoOp	33.41	23.71	27.74
MaPLe	37.44	35.61	36.50
PromptSRC	42.73	37.87	40.15
PromptKD	49.12	41.81	45.17
Δ	+6.39	+3.94	+5.02

(h) FGVCAircraft

ViT-B/16	Base	Novel	HM
CLIP	56.48	64.05	60.03
CoOp	92.19	54.74	68.69
CoCoOp	87.49	60.04	71.21
MaPLe	94.07	73.23	82.35
PromptSRC	92.90	73.90	82.32
PromptKD	97.54	82.08	89.14
Δ	+4.64	+8.18	+6.82

(k) EuroSAT

ViT-B/16	Base	Novel	HM
CLIP	96.84	94.00	95.40
CoOp	98.00	89.81	93.73
CoCoOp	97.96	93.81	95.84
MaPLe	97.74	94.36	96.02
PromptSRC	98.10	94.03	96.02
PromptKD	98.91	96.65	97.77
Δ	+0.81	+2.62	+1.75

(c) Caltech101

ViT-B/16	Base	Novel	HM
CLIP	72.08	77.80	74.83
CoOp	97.60	59.67	74.06
CoCoOp	94.87	71.75	81.71
MaPLe	95.92	72.46	82.56
PromptSRC	98.07	76.50	85.95
PromptKD	99.42	82.62	90.24
Δ	+1.35	+6.12	+4.29

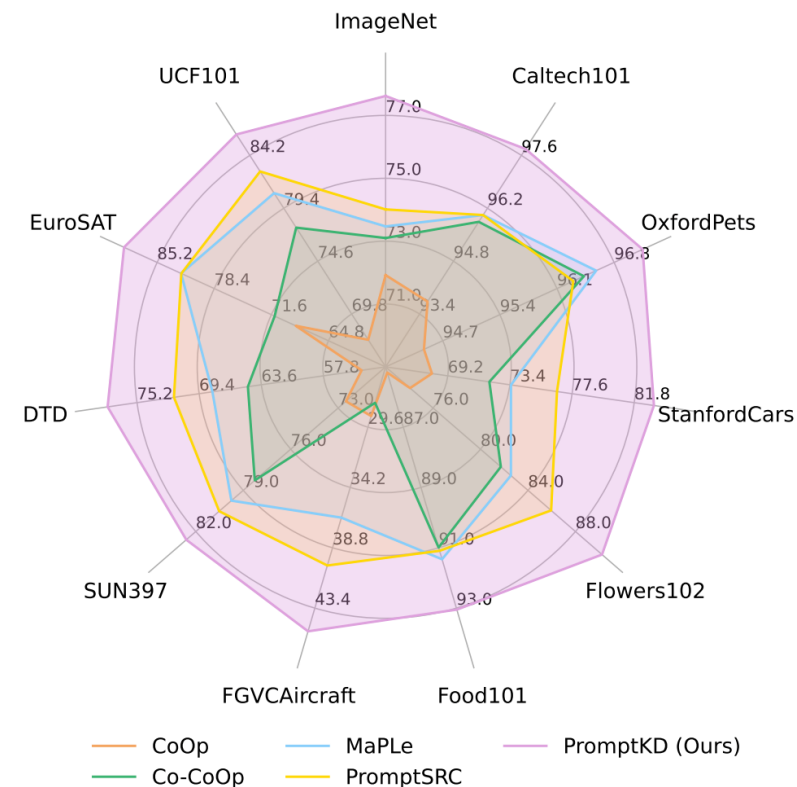
(f) Flowers102

ViT-B/16	Base	Novel	HM
CLIP	69.36	75.35	72.23
CoOp	80.60	65.89	72.51
CoCoOp	79.74	76.86	78.27
MaPLe	80.82	78.70	79.75
PromptSRC	82.67	78.47	80.52
PromptKD	83.69	81.54	82.60
Δ	+1.02	+3.07	+2.08

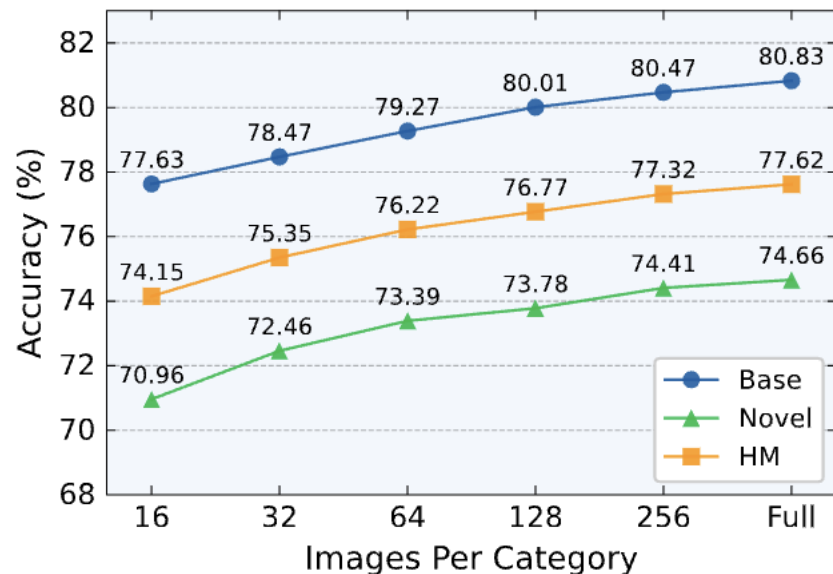
(i) SUN397

ViT-B/16	Base	Novel	HM
CLIP	70.53	77.50	73.85
CoOp	84.69	56.05	67.46
CoCoOp	82.33	73.45	77.64
MaPLe	83.00	78.66	80.77
PromptSRC	87.10	78.80	82.74
PromptKD	89.71	82.27	86.10
Δ	+2.61	+3.47	+3.36

(l) UCF101



Experiments



Improved ImageNet classification accuracy of the student model with increasing numbers of unlabeled images per category used for distillation.

Method	Base	Novel	HM
CLIP	72.43	68.14	70.22
Projector Only	78.48	72.79	75.53
Full Fine-tune	75.90	70.95	73.34
w/o Shared Text Feature	78.79	73.37	75.98
PromptKD	79.27	73.39	76.22

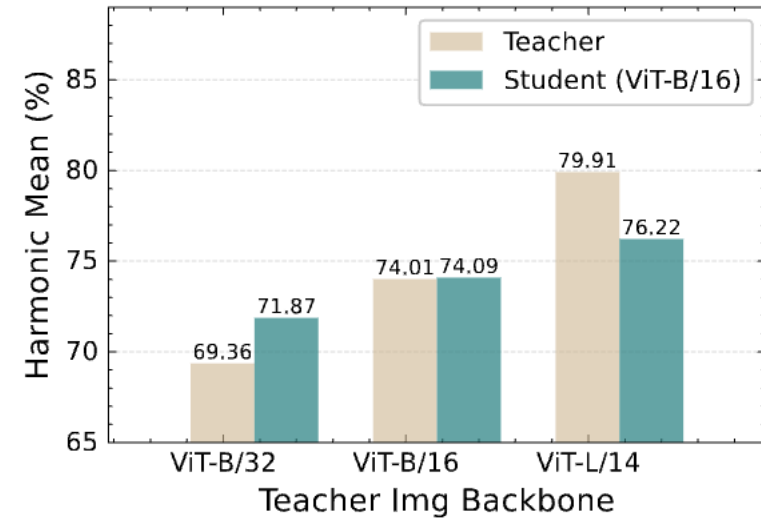
Ablation of different distillation ways.



Experiments

Role (Method)	Img Backbone	Base	Novel	HM
CLIP	ViT-B/16	72.43	68.14	70.22
PromptSRC	ViT-B/16	77.60	70.73	74.01
Teacher (CLIP)	ViT-L/14	79.18	74.03	76.52
Student	ViT-B/16	76.53	72.58	74.50
Teacher (MaPLe)	ViT-L/14	82.79	76.88	79.73
Student	ViT-B/16	78.43	73.61	75.95
Teacher (PromptSRC)	ViT-L/14	83.24	76.83	79.91
Student	ViT-B/16	79.27	73.39	76.22

Comparison of different pre-training methods.
Teacher pre-training with PromptSRC brings the best student performance



Comparison of distillation results for teachers with different capacities. Better teachers lead to better distillation performance.



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MEGVII 旷视

Cascade Prompt Learning for Visual-Language Model Adaptation

Ge Wu, Xin Zhang, **Zheng Li**, Zhaowei Chen, Jiajun Liang, Jian Yang, Xiang Li

ECCV 2024

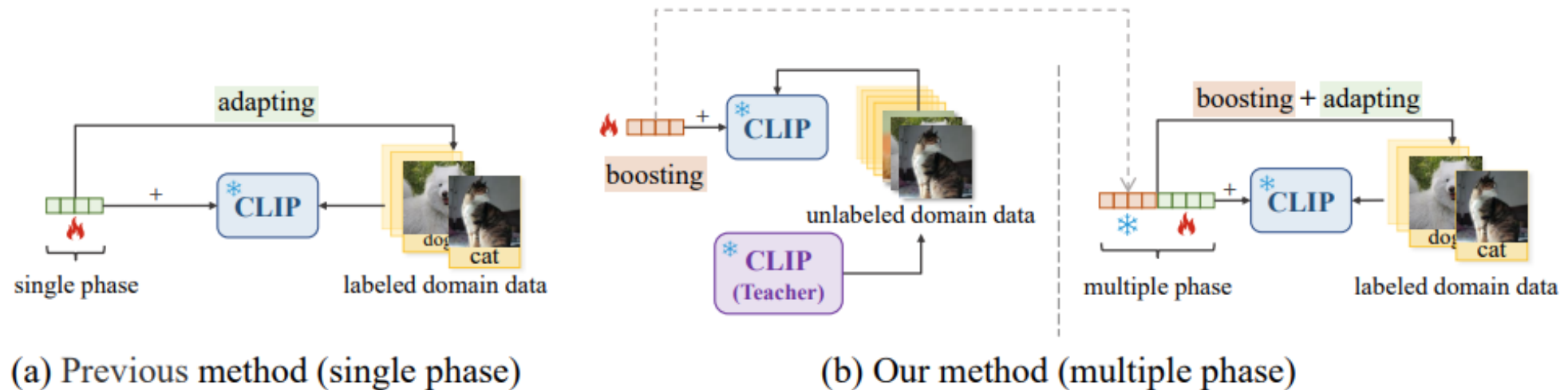
Paper: <https://arxiv.org/abs/2409.17805>

Code: <https://github.com/megvii-research/CasPL>

中文解读: <https://zhuanlan.zhihu.com/p/867291664>



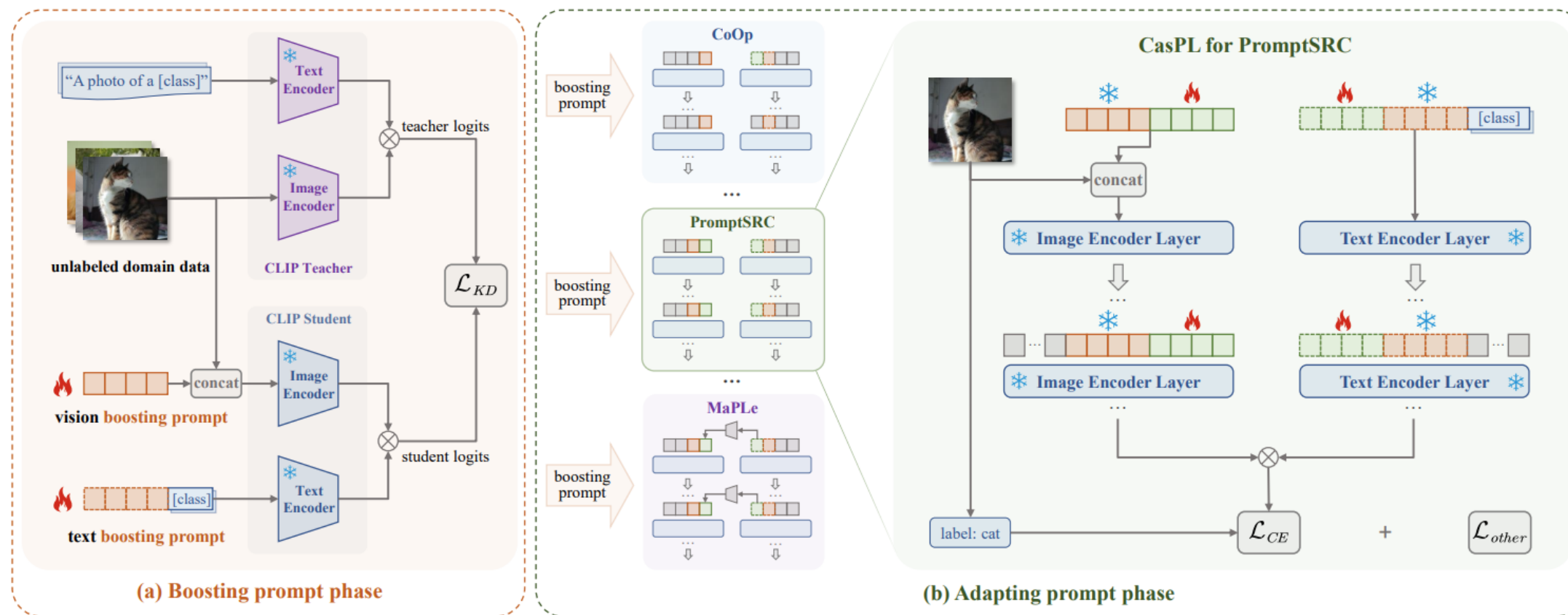
[ECCV 24] CasPL



- Previous methods adopt single phase prompting methods for adapting domain datasets.
- CasPL introduces cascaded diverse prompts with multiple functions consisting of both boosting and adapting prompt phases.



[ECCV 24] CasPL



🔥 Trainable Parameters ❄️ Frozen Parameters ⊗ Multiplication 🟢 Image adapting prompt 🟡 Text adapting prompt 🟠 Image boosting prompt 🟤 Text boosting prompt 🟡 Unlearnable token



SOTA performance

Method	Base	Novel	HM
CLIP	69.34	74.22	71.70
CoOp	82.69	63.22	71.66
+CasPL	84.78	74.49	79.30 (+7.64)
CoCoOp	80.47	71.69	75.83
+CasPL	83.63	78.12	80.78 (+4.95)
MaPLe	82.28	75.14	78.55
+CasPL	84.48	79.59	81.96 (+3.41)
PromptSRC	84.26	76.10	79.97
+CasPL	86.11	79.54	82.69 (+2.72)

(a) Average over 11 datasets.

Method	Base	Novel	HM
CLIP	72.43	68.14	70.22
CoOp	76.47	67.88	71.92
+CasPL	77.90	67.43	72.29 (+0.37)
CoCoOp	75.98	70.43	73.10
+CasPL	77.40	71.40	74.28 (+1.18)
MaPLe	76.66	70.54	73.47
+CasPL	78.20	71.47	74.68 (+1.21)
PromptSRC	77.60	70.73	74.01
+CasPL	78.97	70.50	74.50 (+0.49)

(b) ImageNet

Method	Base	Novel	HM
CLIP	96.84	94.00	95.40
CoOp	98.00	89.81	93.73
+CasPL	98.63	95.50	97.04 (+3.31)
CoCoOp	97.96	93.81	95.84
+CasPL	98.60	94.63	96.58 (+0.74)
MaPLe	97.74	94.36	96.02
+CasPL	98.47	96.03	97.23 (+1.21)
PromptSRC	98.10	94.03	96.02
+CasPL	98.60	95.70	97.13 (+1.11)

(c) Caltech101

Method	Base	Novel	HM
CLIP	91.17	97.26	94.12
CoOp	93.67	95.29	94.47
+CasPL	94.83	97.37	96.08 (+1.61)
CoCoOp	95.20	97.69	96.43
+CasPL	95.23	98.10	96.65 (+0.22)
MaPLe	95.43	97.76	96.58
+CasPL	95.37	98.13	96.73 (+0.15)
PromptSRC	95.33	97.30	96.30
+CasPL	95.53	98.07	96.78 (+0.48)

(d) OxfordPets

Method	Base	Novel	HM
CLIP	63.37	74.89	68.65
CoOp	78.12	60.40	68.13
+CasPL	80.23	75.77	77.94 (+9.81)
CoCoOp	70.49	73.59	72.01
+CasPL	76.37	81.07	78.65 (+6.64)
MaPLe	72.94	74.00	73.47
+CasPL	78.10	81.97	79.99 (+6.52)
PromptSRC	78.27	74.97	76.58
+CasPL	82.33	79.93	81.11 (+4.53)

(e) StanfordCars

Method	Base	Novel	HM
CLIP	72.08	77.80	74.83
CoOp	97.60	59.67	74.06
+CasPL	98.27	73.47	84.08 (+10.02)
CoCoOp	94.87	71.75	81.71
+CasPL	96.63	76.50	85.40 (+3.69)
MaPLe	95.92	72.46	82.56
+CasPL	97.73	77.63	86.53 (+3.97)
PromptSRC	98.07	76.50	85.95
+CasPL	98.73	80.13	88.46 (+2.51)

(f) Flowers102

Method	Base	Novel	HM
CLIP	90.10	91.22	90.66
CoOp	88.33	82.26	85.19
+CasPL	90.60	91.03	90.82 (+5.63)
CoCoOp	90.70	91.29	90.99
+CasPL	91.50	92.93	92.21 (+1.22)
MaPLe	90.71	92.05	91.38
+CasPL	91.43	92.93	92.18 (+0.80)
PromptSRC	90.67	91.53	91.10
+CasPL	91.27	92.20	91.73 (+0.63)

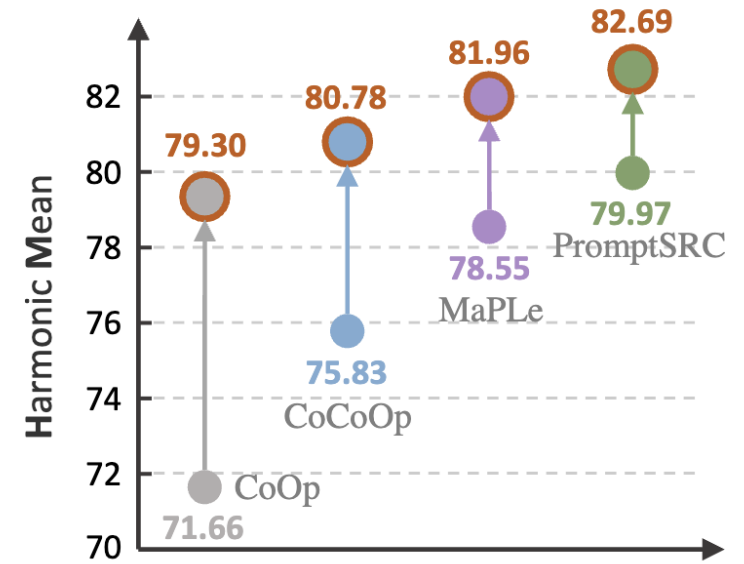
(g) Food101

Method	Base	Novel	HM
CLIP	27.19	36.29	31.09
CoOp	40.44	22.30	28.75
+CasPL	45.83	37.30	41.13 (+12.38)
CoCoOp	33.41	23.71	27.74
+CasPL	42.93	41.23	42.07 (+14.33)
MaPLe	37.44	35.61	36.50
+CasPL	43.60	42.20	42.89 (+6.39)
PromptSRC	42.73	37.87	40.15
+CasPL	48.23	41.97	44.88 (+4.73)

(h) FGVCAircraft

Method	Base	Novel	HM
CLIP	69.36	75.35	72.23
CoOp	80.60	65.89	72.51
+CasPL	81.77	72.77	77.00 (+4.49)
CoCoOp	79.74	76.86	78.27
+CasPL	80.20	79.10	79.65 (+1.38)
MaPLe	80.82	78.70	79.75
+CasPL	82.23	79.80	81.00 (+1.25)
PromptSRC	82.67	78.47	80.52
+CasPL	83.10	79.53	81.28 (+0.76)

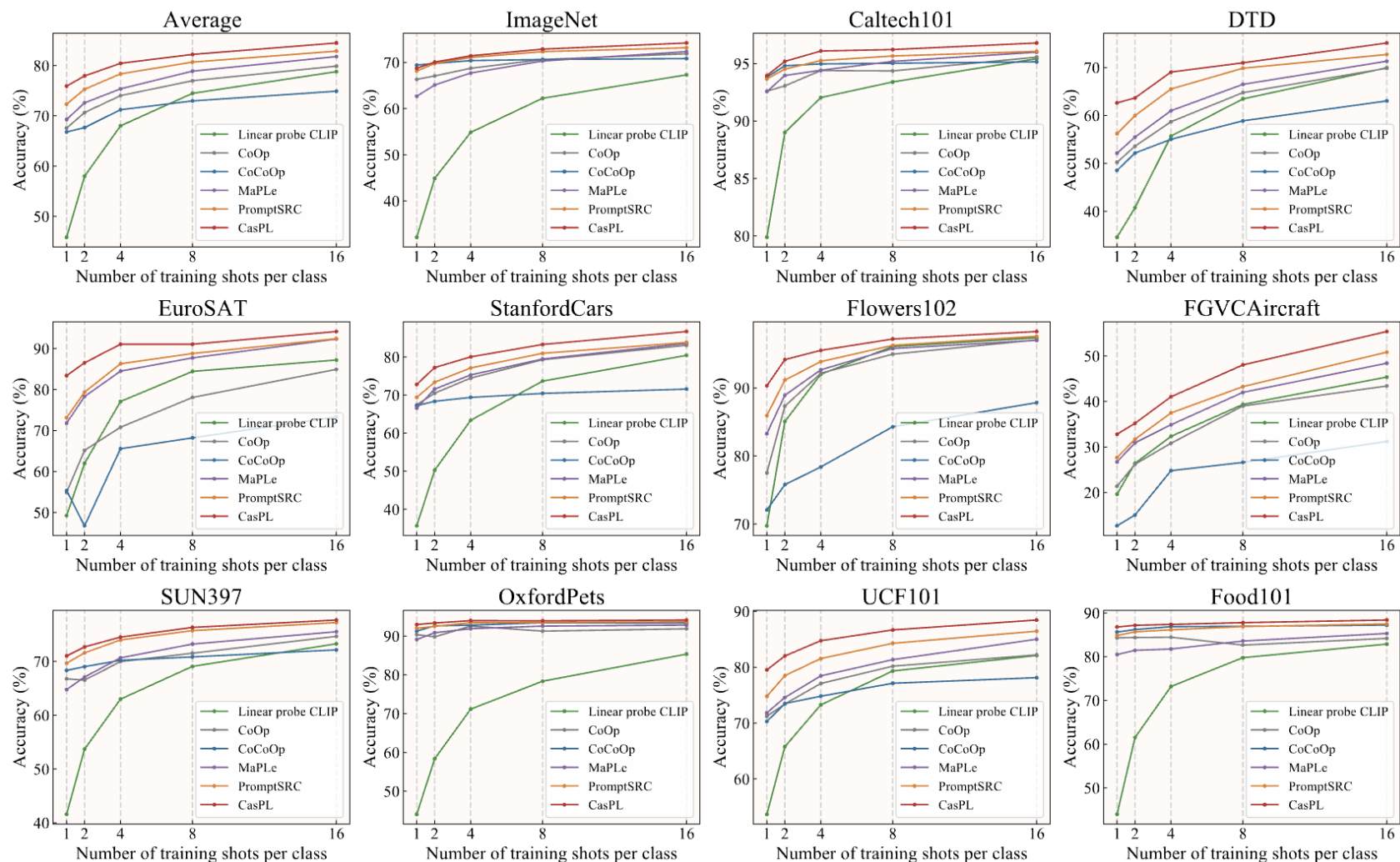
(i) SUN397



Performance w/ and w/o CasPL.



Experiments



CasPL performance comparison in a few-shot image recognition setting.



References

- Zhou et al. Learning to prompt for vision-language models. IJCV 2022.
- Zhou et al. Conditional prompt learning for vision-language models. CVPR 2022.
- Khattak et al. Maple: Multi-modal prompt learning. CVPR 2023.
- Yao et al. Visual-language prompt tuning with knowledge-guided context optimization. CVPR 2023
- Zhu et al. Prompt-aligned gradient for prompt tuning. ICCV 2023.
- Khattak et al. Self-regulating prompts: Foundational model adaptation without forgetting. ICCV 2023.
- Ren et al. Prompt pre-training with twenty-thousand classes for open-vocabulary visual recognition. NeurIPS 2023.
- Roy et al. Consistency-guided prompt learning for vision-language models. ICLR 2024.
- Li et al. Promptkd: Unsupervised prompt distillation for vision-language models. CVPR 2024.
- Wu et al. Cascade Prompt Learning for Vision-Language Model Adaptation. ECCV 2024.



感谢观看！

如果大家有任何想要交流的问题，欢迎在下方交流区留言~

